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Sex differences in body composition and shock attenuation during running

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A B S T R A C T

Running-related impact shock is absorbed via biological tissue deformation. Given known sex differences in body composition, shock attenuation may also differ between sexes thereby influencing sex-specific running-related injury risk. This study examined sex differences in body composition and shock attenuation during running. Seventeen female (mean ± 1SD age: 34.7 ± 16.1) and twenty-one male runners (age: 29.0 ± 13.8) ran overground as inertial measurement units with triaxial accelerometers measured impact shock at the distal tibia and low-back. Frequency-domain axial and resultant shock attenuation were calculated between the low-back relative to the tibia using a transfer function of the power spectral density within 9–20, 21–35, and 36–50 Hz. Bone mineral density and content, fat and lean mass were measured in the lower extremity and pelvis/gynoid regions using dual x-ray absorptiometry. The association between sex and shock attenuation was tested using age-adjusted linear regression models, adjusted and unadjusted for body composition as a post-hoc analysis ($\alpha = 0.05$). Body composition variables normalized to body mass were compared between sexes using independent samples t-tests ($\alpha = 0.05$). Body composition differed between sexes (p-range: <0.001–0.01, Cohen's d range: 0.17–2.41). Before adjusting for body composition, sex was not significantly associated with axial or resultant shock attenuation ($p > 0.05$), but adjusting for select body composition variables like lower extremity lean and bone mass revealed greater attenuation in females than males (β-range: −124.76 to −46.42, negative indicates greater attenuation; p-range = 0.004–0.04). Sex may not influence shock attenuation during running, but body composition must be accounted for to better understand this association and consequently sex-specific tissue capacities relative to applied loads.

1. Introduction

Running-related injuries are sex-specific (Hollander et al., 2021). Bone stress injuries, for instance, are more prevalent in female than male runners (Hoenig et al., 2023) and lead to considerable training time loss. While the magnitude of running-related external loads may not dictate injury risk (Gruber, 2023), the body's attenuation response to impact shock, for example, may reflect the management of external loads by musculoskeletal tissues and therefore their potential for injury. Consequently, sex-specific differences in body composition and attenuation responses could contribute to sex-specific running-related injury rates.

Foot-ground contact generates an impact shock, accelerating body segments during each running step with frequencies between 3–50 Hz (Shorten & Winslow, 1992). Examining these frequencies at the tibia relative to the low-back or head reveals the degree of shock attenuated by the body (Edwards et al., 2012; Gruber et al., 2014; Lafortune et al., 1996). While shock between 3–8 Hz may be attenuated actively by eccentric muscle actions (Derrick et al., 1998; Edwards et al., 2012), studies in animals, cadavers, and live humans suggests passive attenuation of higher frequency shock between 9–50 Hz occurs through bone, fat, and lean tissue deformation (Chu et al., 1986; Nigg et al., 1995; Paul et al., 1978; Radin et al., 1970; Schinkel-Ivy et al., 2012). Consequently, passive shock attenuation demands met through tissue deformation may influence running-related injury risk depending on musculoskeletal tissue size and composition. For instance, female runners may have a greater injury risk given that their lower bone mass and smaller bone geometry (Kelsey et al., 2007; Meardon et al., 2021) will result in larger peak bone stresses and strain volume compared with male runners (Bruce & Edwards, 2023; Meardon et al., 2021). However, it is unknown whether passive shock attenuation during running, achieved via tissue deformation, differs between males and females who commonly differ in body composition (Hunter et al., 2023).

Distinct shock attenuation profiles for males and females may indicate differing capacities to manage external loads and running-related injury risks. Female runners tend to exhibit lower bone mineral density in adolescence (Kelsey et al., 2007), and smaller bone geometry with more fat mass in adulthood than males (Giandolini et al., 2019; Meardon et al., 2021). Bone, fat, and lean tissue enable shock attenuation given their viscoelastic nature (Schinkel-Ivy et al., 2012). Consequently, female runners may rely more on fat or lean mass than bone to attenuate

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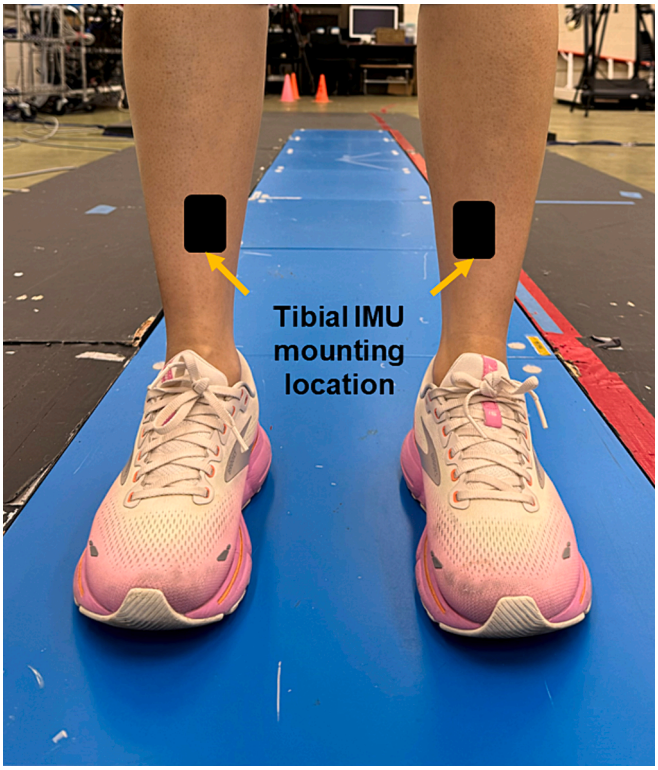


Fig. 1. The IMUs are visualized at the antero-medial distal tibia where they were affixed for data collection. The IMUs were firmly secured at these locations with double sided tape and elastic bandages.

high-frequency (9–50 Hz) shock compared with males, due to smaller bone mass and geometry in females. Indeed, female runners experience greater bone tissue deformation than males (Bruce & Edwards, 2023; Meardon et al., 2021), which may reflect an altered reliance on bone, lean, and fat mass between sexes. These sex differences in body composition and shock attenuation may contribute to bone stress injuries being more common in females than males (Hollander et al., 2021).

Therefore, this study aimed to identify the association between sex and shock attenuation of impact shock frequencies 9–20 Hz (FB1), 21–35 Hz (FB2) and 36–50 Hz (FB3) along both axial and resultant components. The axial component has been primarily analyzed in relation to running injuries (Milner et al., 2006; Popp et al., 2023). However, the anteroposterior and mediolateral components of impact shock are also substantial (LaFortune, 1991), leading to recommendations to analyze the resultant of running-related impact shock (Chadefaux et al., 2019; Sheerin et al., 2019) as done in the present study. This study also aimed to analyze differences in body-composition between males and females which may influence the associations between sex and shock attenuation. Given that female runners tend to have less lean mass (Giandolini et al., 2019) and smaller bone geometry (Bruce & Edwards, 2023; Meardon et al., 2021) than males, they may attenuate less shock than males. Conversely, their typically greater fat mass than males (Giandolini et al., 2019) may compensate for the lesser degree of shock attenuated through bone and lean mass deformation. Therefore, it was hypothesized that shock attenuation across frequency bands would be similar in female and male runners.

2. Methods

2.1. Subject characteristics

Body composition data from Giandolini et al. (2019) and shock attenuation data from Sinclair (2016) were used in a-priori power calculations (G*Power Version 3.1.9.4), revealing 36 subjects were required to detect a relationship between sex and shock attenuation with a Cohen's d effect size: 0.98, 80 % statistical power, at nominal significance level equal to 0.05. Therefore, 38 runners (17 female, 21 male) were recruited.

Subjects had to be 18–65 years old, with a weekly running volume of 3–30 miles•week^{-1}, no history of lower-back or limb orthopedic surgery in the previous 10-years, and free of chronic illnesses that may restrict engagement in low-to-moderate intensity physical activity. If potential subjects experienced a musculoskeletal injury in the past 6-months, they must have resumed normal running training for $\geq$ 8 weeks prior to enrollment. Individuals who used tobacco, had a history of pulmonary or cardiovascular disease, neurological dysfunction, and/or current pregnancy were excluded. All subjects gave written informed consent to participate. The study was approved by the university's Institutional Review Board.

2.2. Gait analysis procedures

Subjects completed a modified Physical Activity Readiness Questionnaire (Riebe, 2015) to confirm health status before testing. Age, biological sex, current weekly running volume (km), and running experience (years) were obtained through questionnaires.

Subjects wore form-fitting clothes and laboratory running shoes (Reebok; One X CrossFit Cushion 3.0) to standardize footwear effects on shock attenuation. Height and weight were measured, and then subjects performed a 5–8 min treadmill warm-up at a self-selected running speed with a perceived exertion of $\leq$ 11 (Borg, 1998). Next, inertial measurement units (IMUs) with triaxial accelerometers (Blue Trident, IMeasureU, Auckland, NZ; sampling frequency: 1600 Hz, capture range: $\pm$200 g, weight: 12 g) were affixed bilaterally to the skin over the antero-medial distal tibia (Fig. 1) and low-back (lumbosacral joint) (Wosk & Voloshin, 1981). IMUs were firmly secured with double-sided tape and elastic bandages. Consistent with previous research (Gruber et al., 2014; Reenalda et al., 2019; Van den Berghe et al., 2019), IMUs were oriented so that the vertical axis of the IMU was aligned in the axial direction of the tibia. The direction of the anteroposterior and mediolateral axes of the IMU were dependent on the shape of the tibia. Each IMU was calibrated following manufacturer guidelines.

Subjects performed five practice runs at the target running speed (3.35 m•s^{-1} $\pm$ 5 %) across an 18 m level-ground platform embedded with three force plates (OR-6–2000; AMTI Inc, Watertown, MA, USA) to acclimate to the speed and lab environment. A standardized speed was used to control for speed-effects on shock attenuation (Mercer et al., 2002). Force plate data were collected at 1200 Hz using Qualisys Track Manager (QTM; Qualisys AB, Göthenburg, SWE). Each data collection trial began with subjects performing a jump then they immediately commenced running at the target speed. The jump was manually flagged in QTM which served to align the IMU and force plate data to identify stance phases for analyses. Two optoelectrical timing gates (TC-Gait, Brower Timing System, Draper, UT, USA) were placed 6 m apart to monitor running speed over the force plates. Eight successful trials of each limb were retained for analyses. A trial was successful if the subject ran within $\pm$ 5 % of the target running speed, maintained their speed between timing gates, made $\geq$ 1 force plate contact with their entire foot, and did not intentionally target the force plates. Subjects were instructed to avoid targeting the force plates.

2.3. Body composition measurements

While all subjects underwent both gait analysis and body composition measurements, these procedures were conducted on separate days for 13 out of the 38 total subjects (mean $\pm$ SD=10.8 $\pm$ 9.1 days apart) due to subject availability. For these individuals, a hand-to-leg bioelectrical impedance analysis (BIA; Tanita, MC-780U plus P)

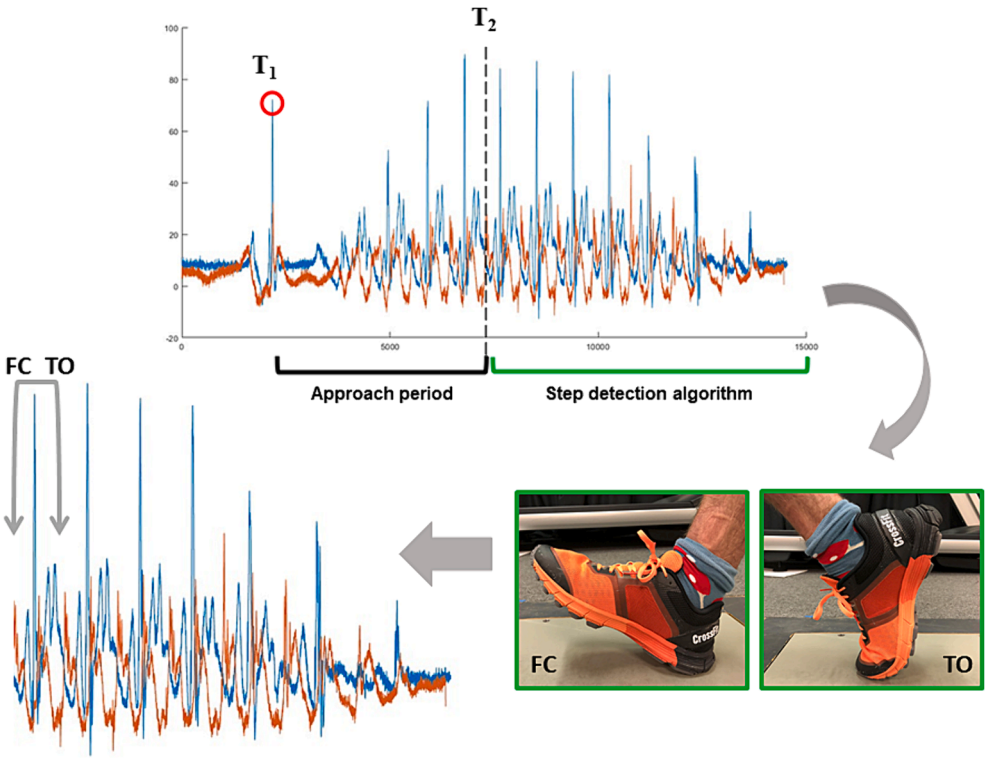


Fig. 2. Workflow to identify the stance phase of the same step recorded by both force plate and IMU data, which ensured the analyzed step was performed at a constant, standardized speed. The time elapsed between the jump landing at T_1 and foot-contact (FC) on a force plate at T_2 was called the approach period. The acceleration signal between T_1 and T_2 was entered into the step detection algorithm to confirm the instant of the FC gait event. Data from each acceleration component between FC and toe off (TO) were retained for frequency analysis.

confirmed consistent body composition across visits.

Lower extremity and pelvis/gynoid bone mineral density (BMD), bone mineral content (BMC), fat mass, and lean mass were measured via full-body dual x-ray absorptiometry (iDXA; Lunar Prodigy, enCORE version 17, GE Medical Systems, Madison, WI, USA) while subjects lay supine.

2.4. Data processing

Acceleration data from the low-back and tibia IMUs were converted to units of gravity in MATLAB (Release 2022a, MathWorks, Natick, MA, USA). The signal from each axis was detrended by subtracting a least-squares best-fit line. Upon removing bias, including the 1g measured by the accelerometers at rest, each signal oscillated about zero. Individual stance phases were subsequently detected from each running trial for the frequency analysis.

2.5. Stance-phase detection

Despite being initiated simultaneously, each IMU sensor began collecting data at slightly different times with sampling frequencies varying in a small range (right leg: 1598.15 Hz, left leg: 1604.32 Hz, low-back: 1587.43 Hz). To address these disparities, the first Unix timestamp common to both the tibia and low-back signals was identified as the start of each signal, ensuring temporal alignment then the "resample" function in MATLAB was used to linearly interpolate each signal to 1600 Hz.

IMUs recorded all steps completed during each trial, including the steps to reach the target running speed. Therefore, the running step common to both the acceleration signal and a force plate contact was identified for analysis (Fig. 2) because the force plates were located between the timing gates ensuring target running speed. After aligning the tibial and low-back acceleration signals, this common step was identified by first determining the instant of the jump landing performed at the beginning of each trial from the tibial IMU data as the instant of the first acceleration peak exceeding 3 g (T_1). Next, T_1 was temporally aligned with the manually flagged jump landing recorded in the force plate timeseries, thereby synchronizing the IMU data with the force plates (refer to Supplemental Material 1 for the analysis of lag or lead times of the manual flag relative to T_1). Then, the temporal location of foot-contact within the force plate timeseries was identified as the instant when the vertical ground reaction force exceeded 20 N. The time interval in seconds between the jump landing (T_1) and foot-contact on the force-plate (T_2) was defined as the 'approach period'. The tibial acceleration peak which immediately followed T_2 indicated the stance phase within the tibial and low-back acceleration signals that temporally aligned with the force plate data. Given the subjectivity of the manual flag, the desired instant of foot-contact within the acceleration signals was confirmed using a validated algorithm (Aubol & Milner, 2020). Toe-off within the acceleration signals was calculated using the force-plate derived contact time. Contact time was the time between when the ground reaction force exceeded (foot-contact) and subsequently fell below (toe-off) 20 N. The instants of foot-contact and toe-off were then applied to the unfiltered tibial and lower back acceleration signals to isolate the stance phases used for the frequency-domain analyses.

2.6. Frequency domain analyses

Individual stance phases of the unfiltered axial and resultant acceleration signals from tibia and lower-back were analyzed in the frequency domain. The resultant acceleration component was calculated as $\sqrt{x^2+y^2+z^2}$. Resultant and axial signals were zero-padded at both ends (Shorten and Winslow, 1992). Signal power was calculated using the power spectral density (PSD) with the Hanning window. Signal power for frequencies between 0 Hz and 0.5*sampling frequency was normalized to 1 Hz bins (Edwards et al., 2012; Gruber et al., 2014; Hamill et al., 1995). The change in signal power between the tibia and

Table 1
Anthropometrics and demographics for both male and female runners. * indicates statistically significant at $p \leq 0.05$.

	Female mean ± SD	Male mean ± SD	p-value	Cohen's D
Sample size	17	21	–	–
Age (years)	34.7 ± 16.1	29.0 ± 13.8	0.39	0.38
Height (meters)	1.64 ± 0.05	1.77 ± 0.04	<0.001*	2.62
Body Mass (kg)	60.0 ± 7.1	75.1 ± 7.1	<0.001*	2.10
Years Run	12.4 ± 12.0	9.3 ± 12.7	0.18	0.25
Weekly volume (kms)	22.5 ± 8.8	20.2 ± 10.2	0.47	0.23

Table 2
Body composition variables for female and male runners. * indicates statistically significant at $\alpha = 0.05$.

	Female mean ± SD	Male mean ± SD	p-value	Cohen's D
Lower extremity fat mass (grams/kg)	56.89 ± 9.58	35.40 ± 8.14	<0.001*	2.41
Lower extremity lean mass (grams/kg)	112.70 ± 10.21	129.20 ± 10.62	<0.001*	1.58
Lower extremity BMC (grams/kg)	7.13 ± 0.64	7.77 ± 0.83	0.01*	0.86
Lower extremity BMD (grams/cm^2)	1.20 ± 0.09	1.39 ± 0.10	<0.001*	0.24
Pelvis BMD (grams/cm^2)	1.02 ± 0.09	1.16 ± 0.15	0.001*	0.17
Pelvis BMC (grams/kg)	5.02 ± 0.59	5.30 ± 0.69	0.18	0.43
Gynoid Fat mass (grams/kg)	57.58 ± 9.75	34.06 ± 10.76	<0.001*	2.29
Gynoid Lean mass (grams/kg)	103.39 ± 8.12	117.77 ± 12.63	<0.001*	1.35

lower-back within each bin was calculated with a transfer function:

$$10 \times \log_{10}\left(\frac{PSD_{low-back}}{PSD_{tibia}}\right)$$

where negative values indicate signal power attenuation between the tibia (PSD_{tibia}) and lower-back ($PSD_{low-back}$) whereas positive values indicate a gain in signal power. The magnitude of shock attenuation (or gain) was calculated as the integral of the transfer function result within the: 9–20 Hz (FB1), 21–35 Hz (FB2) and 36–50 Hz (FB3) frequency bands. FB1 has been previously analyzed in running as the "high-frequency" band (Gruber et al., 2014). FB2 and FB3 were also analyzed because frequency components relevant to running impacts can be as high as 50 Hz (Shorten & Winslow, 1992). Impact shock signal power magnitude was also computed as the integral of the tibial signal power across the FB1, FB2, and FB3 frequency bands. Impact shock signal power was used to identify differences in the input signal at the tibia between sexes which aided in the interpretation of sex differences in shock attenuation.

2.7. iDXA measurements

The following body composition measures were exported from the iDXA software: lower extremity areal BMD, BMC, lean mass, fat mass; pelvis areal BMD; and gynoid lean and fat mass. Each mass-related body composition variable was divided by body mass for normalization purposes.

2.8. Statistical analysis

All statistical analyses were performed in R (version R-4.1.3). Normality was assessed via the Shapiro-Wilk test ($\alpha = 0.05$). Normality assumption was not rejected for any of the body composition variables considered. However, signal power magnitude and axial and resultant shock attenuation were not normally distributed.

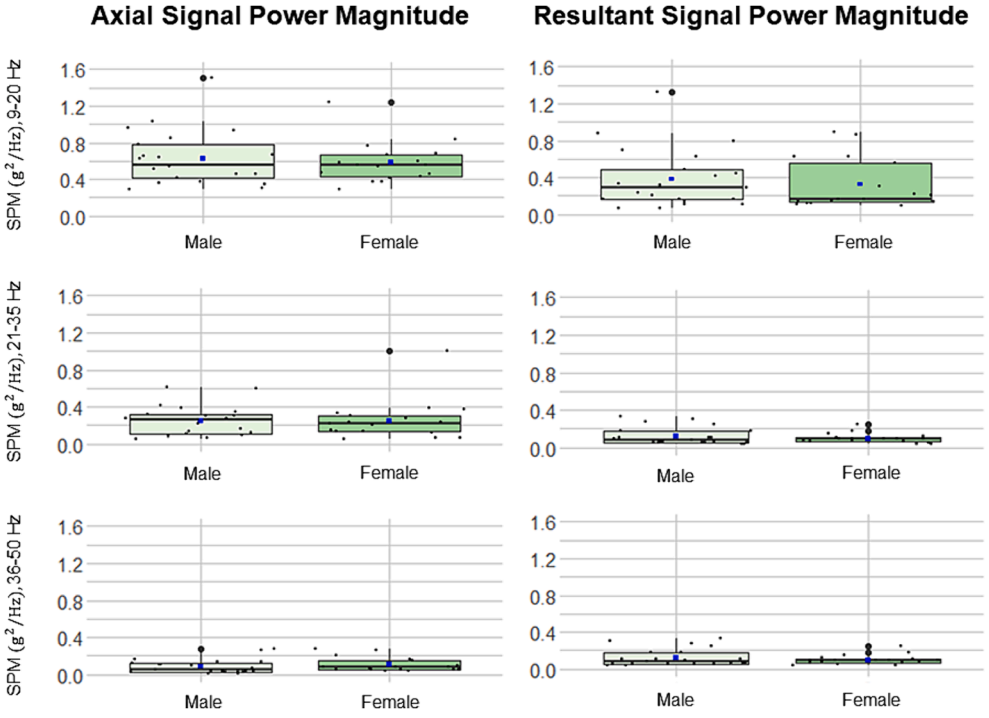


Fig. 3. Axial (right column) and resultant (left column) signal power magnitude (SPM) at the tibia for male and female runners across the three frequency bands examined (rows). The blue dot indicates the group average, the thick horizontal black line indicates the median. The smaller black dots are individual subject data points, while the bigger black dots are outliers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Linear regression results for the age-adjusted association (original model) between sex and frequency-domain shock attenuation along the axial acceleration component, and the original model further adjusted for body composition. Positive beta coefficients indicate less shock attenuation in females compared with males, and negative beta coefficients indicate greater shock attenuation in females compared with males; *statistically significant at $\alpha = 0.05$.

Axial Attenuation	β-coefficient	95 % CI	p-value
Original model			
FB1 (9–20 Hz)	7.030	−17.066 – 31.126	0.55
FB2 (21–35 Hz)	−28.841	−66.400 – 8.719	0.12
FB3 (36–50 Hz)	−40.324	−95.213 – 14.564	0.14
Original model adjusted for lower extremity fat mass			
FB1 (9–20 Hz)	−23.666	−59.702 – 12.370	0.19
FB2 (21–35 Hz)	−75.838	−132.156 – −19.521	0.009*
FB3 (36–50 Hz)	−120.553	−200.702 – −40.405	0.004*
Original model adjusted for lower extremity lean mass			
FB1 (9–20 Hz)	−2.425	−32.986 – 28.136	0.87
FB2 (21–35 Hz)	−40.567	−88.473 – 7.337	0.09
FB3 (36–50 Hz)	−79.968	−146.976 – −12.960	0.02*
Original model adjusted for lower extremity BMC			
FB1 (9–20 Hz)	4.222	−21.982 – 30.426	0.745
FB2 (21–35 Hz)	−36.459	−76.880 – 3.961	0.07
FB3 (36–50 Hz)	−58.426	−115.939 – −0.911	0.04*
Original model adjusted for lower extremity BMD			
FB1 (9–20 Hz)	10.213	−22.690 – 43.116	0.53
FB2 (21–35 Hz)	−32.689	−84.004 – 18.626	0.20
FB3 (36–50 Hz)	−25.642	−100.298 – 49.031	0.48
Original model adjusted for pelvis BMC			
FB1 (9–20 Hz)	5.413	−19.088 – 29.914	0.65
FB2 (21–35 Hz)	−33.809	−70.760 – 3.141	0.07
FB3 (36–50 Hz)	−43.773	−99.661 – 12.113	0.12
Original model adjusted for pelvis BMD			
FB1 (9–20 Hz)	8.840	−18.530 – 36.210	0.51
FB2 (21–35 Hz)	−35.938	−78.302 – 6.426	0.09
FB3 (36–50 Hz)	−25.803	−87.209 – 35.603	0.39
Original model adjusted for gynoid fat mass			
FB1 (9–20 Hz)	−20.977	−55.458 – 13.503	0.22
FB2 (21–35 Hz)	−70.161	−124.308 – −16.014	0.01*
FB3 (36–50 Hz)	−124.769	−198.960 – −50.578	0.001*
Original model adjusted for gynoid lean mass			
FB1 (9–20 Hz)	−9.857	−36.695 – 16.980	0.46
FB2 (21–35 Hz)	−48.436	−91.801 – −5.070	0.02*
FB3 (36–50 Hz)	−76.491	−138.215 – −14.768	0.01*

2.9. Bilateral differences

Axial and resultant signal power, shock attenuation, and lower extremity body composition measures were compared between limbs to assess potential asymmetries. Bilateral differences in body composition parameters were tested using repeated measures t-tests and bilateral differences in shock attenuation variables were tested with Wilcoxon Signed-rank tests ($\alpha = 0.05$). No bilateral differences in signal power magnitude, shock attenuation, or body composition were revealed ($p > 0.05$). Hence, these variables were averaged across limbs for each subject for further analyses.

2.10. Association between sex and shock attenuation

The associations between sex and axial and resultant shock attenuation within FB1–FB3 were tested using linear regression models adjusted for age due to its potential influence on body composition and running mechanics (DeVita et al., 2016; Ponti et al., 2020). Males were the reference group. Shock attenuation contains negative magnitudes, so positive regression coefficients indicated less shock attenuation in females than males, and negative regression coefficients indicated greater shock attenuation in females than males. All linear regression model assumptions and influential outliers were examined. Signal power magnitude within each frequency band was compared between sexes using Wilcoxon Signed-rank tests.

2.11. Sex differences in body composition

Body composition variables, normalized to body mass, were compared between sexes to contextualize the effects of sex on shock attenuation. Body composition variables in the lower extremity and pelvis/gynoid BMD, BMC, fat mass, and lean mass were compared between groups using independent measures t-tests.

2.12. Post-hoc analysis: Effect of body composition on the association between sex and shock attenuation

To better understand the relationship between sex and shock attenuation, each age-adjusted regression model testing the association between sex and shock attenuation (original models) was also adjusted for each body composition variable, yielding 8 (body composition) x 3 (frequency band) x 2 (axial and resultant components) separate models. This was an exploratory analysis because the sample size was insufficient to analyze associations beyond the main effect of sex on shock attenuation. The relationship between body composition and shock attenuation was visualized by sex to aid an understanding of the potential effects of body composition on the association between sex and shock attenuation.

2.13. Manuscript editing tools

ChatGPT 3.5 was utilized for manuscript editing before editing by the senior author, but was not used to generate content.

3. Results

3.1. Anthropometrics and demographics

Female runners were significantly shorter and had less overall body mass than male runners. All other demographic characteristics were similar between sexes (Table 1).

3.2. Sex differences in body composition

Female runners had more fat and less lean mass in the lower extremity and gynoid regions than males (Table 2). Lower extremity BMD and BMC and pelvis BMD were significantly less in females than males (Table 2).

3.3. Association between sex and shock attenuation

Axial and resultant signal power magnitude at the tibia was similar between sexes ($p > 0.05$; Fig. 3). Female runners attenuated significantly more resultant shock within FB3 than males, but the confidence interval was wide ($p = 0.05$; Table 3). Resultant shock attenuation in FB1 and FB2 and axial attenuation across FB1-FB3 were not significantly associated with sex ($p > 0.05$; Fig. 4). All linear regression model assumptions were met, and removing outliers, determined by Cook's distance, did not influence the results therefore outliers remained in the dataset.

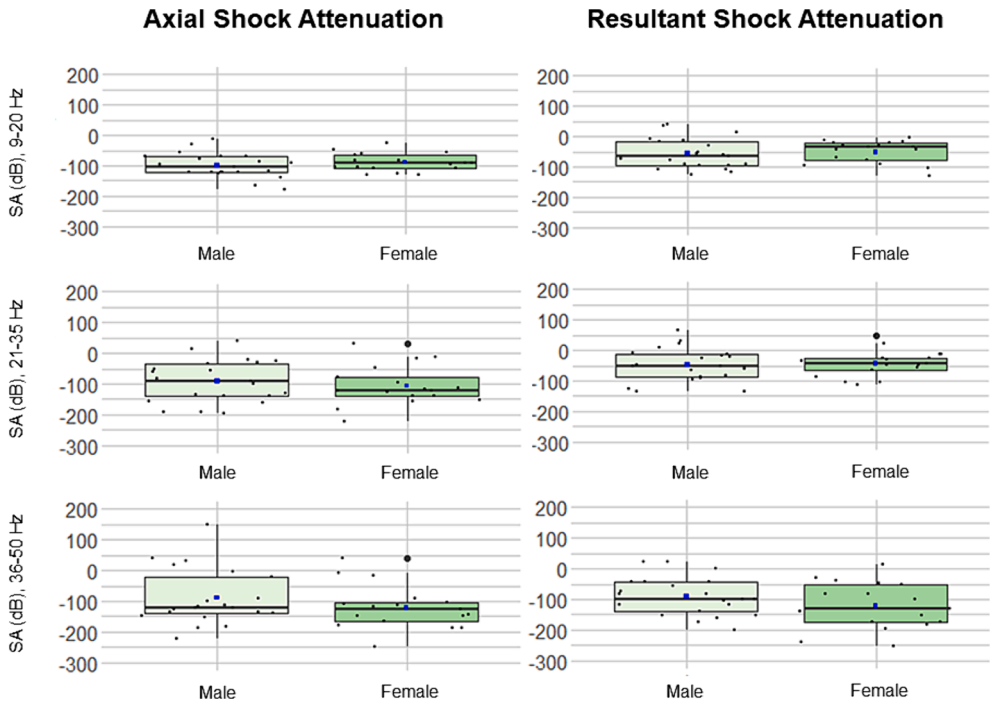


Fig. 4. Axial and resultant shock attenuation (SA) for male and female runners across the three frequency bands examined (rows). The blue dot indicates the group average, while the horizontal line indicates the median. The smaller black dots are individual subject data points, and the bigger black dots are outliers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.4. Post-hoc analysis: Effect of body composition on the association between sex and shock attenuation

Adjusting the original models separately for lower extremity fat mass, BMC, and lean mass revealed females had greater axial shock attenuation than males within specific frequency bands (lower extremity fat mass: FB2, FB3; lower extremity BMC and lower extremity lean mass: FB3; $p < 0.05$, Table 3). Females attenuated more resultant shock in FB3 than males when the original model was separately adjusted for lower extremity BMD, pelvis BMC, and gynoid lean mass ($p < 0.05$, Table 4). Visualization of these significant associations between body composition and shock attenuation stratified by sex (Fig. 5) revealed that lower extremity and gynoid lean mass, pelvis BMC, and lower extremity BMD may influence shock attenuation to different degrees in females compared with males. No other associations between sex and shock attenuation became significant upon adjusting for body composition ($p > 0.05$).

4. Discussion

Lower BMD (Kelsey et al., 2007), smaller bone geometry, and greater fat mass (Giandolini et al., 2019; Meardon et al., 2021) are typical characteristics of female compared with male runners, which suggest the reliance on these tissues to attenuate impact-related shock may vary between sexes. Therefore, this study examined the associations between sex and shock attenuation during overground running, and considered the potential effects of body composition on these associations. When the association between sex and shock attenuation was unadjusted for body composition, male and female runners with comparable tibial shock signal power experienced similar shock attenuation across signal frequencies representing impact, despite significant differences in body composition. While female runners attenuated significantly more shock within 36–50 Hz along the resultant component, this finding must be interpreted cautiously because of its wide confidence interval. Therefore, the hypothesis that shock attenuation would be similar between males and females is supported. Notably, however, a few of the null associations between sex and shock attenuation became significant in the post-hoc analysis whereby female runners attenuated more shock than males upon adjusting for some body composition variables. The results of this post-hoc analysis suggest that the effects of body composition on this association must be further explored with a larger sample size.

Shock attenuation was expected to differ between sexes due to significant differences in body composition because deformation of tissues like bone, articular cartilage, and the heel fat pad were responsible for attenuating impact-related shock in animal models and human pendulum experiments (Chu et al., 1986; Paul et al., 1978; Schinkel-Ivy et al., 2012). Shock attenuation was similar between sexes despite females having smaller bone and lean masses, but greater fat mass, than males. However, the post-hoc analysis (Table 4) revealed females attenuated more shock than males after accounting for lower extremity fat mass, BMC, and lean mass. These results lead to the hypothesis that bones and lean tissues of female runners may undergo significantly more deformation to attenuate similar, if not more, shock as male runners. This hypothesis is supported by evidence that women, despite having smaller bone geometry, do not experience proportionally smaller tibial forces than men during walking and running (Meardon et al., 2021). Consequently, for each step of running, the stress experienced by bone, and possibly lean mass, due to external mechanical loading will likely be greater for females than males. Thus, neglecting sex differences in body composition when prescribing standardized training loads to runners may lead to lower tissue capacities relative to applied loads in females, elevating their risk of running-related injuries such as stress fractures (Popp et al., 2017).

Based on the post-hoc analysis performed in this study, select body composition variables may be related to shock attenuation differently in female compared with male runners. For example, the differences in slopes in Fig. 5 suggest resultant shock attenuation within 36–50 Hz may increase by a greater degree with every unit increase in pelvis BMC for female than male runners. Although a sex-stratified statistical analysis was not feasible due to the small sample size, these exploratory, post-hoc findings suggest a need to investigate whether the effect of body

Table 4
Linear regression results for the age-adjusted association (original model) between sex and frequency-domain shock attenuation along the resultant acceleration component, and the original model further adjusted for body composition. Positive beta coefficients indicate less shock attenuation in females compared with males, and negative beta coefficients indicate a greater degree of shock attenuation in females compared to males; *statistically significant at $p \leq 0.05$.

Resultant Attenuation	β-coefficient	95 % CI	p-value
Original model			
FB1 (9–20 Hz)	0.821	−27.874 – 29.517	0.95
FB2 (21–35 Hz)	−2.016	−35.087 – 31.055	0.90
FB3 (36–50 Hz)	−42.595	−85.423 – 0.233	0.05*
Original model adjusted for lower extremity fat mass			
FB1 (9–20 Hz)	4.320	−41.620 – 50.261	0.84
FB2 (21–35 Hz)	−9.763	−62.626 – 43.099	0.70
FB3 (36–50 Hz)	−59.743	−127.914 – 8.428	0.08
Original model adjusted for lower extremity lean mass			
FB1 (9–20 Hz)	−29.410	−39.739 – 34.042	0.52
FB2 (21–35 Hz)	−6.453	−48.963 – 36.057	0.75
FB3 (36–50 Hz)	−49.750	−104.749 – 5.248	0.07
Original model adjusted for lower extremity BMC			
FB1 (9–20 Hz)	−6.757	−37.296 – 23.781	0.65
FB2 (21–35 Hz)	−10.087	−45.424 – 25.250	0.56
FB3 (36–50 Hz)	−56.556	−101.478 – −11.633	0.01*
Original model adjusted for lower extremity BMD			
FB1 (9–20 Hz)	6.548	–32.571 – 45.668	0.73
FB2 (21–35 Hz)	4.550	−40.537 – 49.637	0.83
FB3 (36–50 Hz)	−59.900	−117.759 – −2.040	0.04*
Original model adjusted for pelvis BMC			
FB1 (9–20 Hz)	2.291	−27.021 – 31.604	0.87
FB2 (21–35 Hz)	−1.994	−35.993 – 32.003	0.90
FB3 (36–50 Hz)	−46.526	−89.652 – −3.399	0.03*
Original model adjusted for pelvis BMD			
FB1 (9–20 Hz)	6.231	−26.135 – 38.598	0.69
FB2 (21–35 Hz)	6.987	−29.974 – 43.949	0.70
FB3 (36–50 Hz)	−46.421	−95.042 – 2.199	0.06
Original model adjusted for gynoid fat mass			
FB1 (9–20 Hz)	1.308	−42.595 – 45.213	0.95
FB2 (21–35 Hz)	−19.172	−69.137 – 30.792	0.44
FB3 (36–50 Hz)	−56.869	−122.060 – 8.320	0.08
Original model adjusted for gynoid lean mass			
FB1 (9–20 Hz)	−3.548	−37.961 – 30.863	0.83
FB2 (21–35 Hz)	−16.806	−55.423 – 21.810	0.38
FB3 (36–50 Hz)	−63.831	−113.484 – −14.178	0.01*

composition on shock attenuation varies by sex in a larger sample.

Our hypotheses assume the proportion of passive shock attenuation across frequency bands in this study was similar between males and females and do not account for the role of active shock attenuation mechanisms. Active mechanisms of the lower extremity can attenuate frequencies as high as 35 Hz (Edwards et al., 2012). These mechanisms may include, but are not limited to, sagittal plane knee kinematics and joint stiffness (Edwards et al., 2012; Hamill et al., 2014; Lafortune et al., 1996) which may or may not differ between sexes (Sinclair et al., 2015; Almonroeder & Benson, 2017; Ferber et al., 2003). Thus, future research must explore if the relationship between active mechanisms and the attenuation of specific frequencies differs between sexes.

The post-hoc analysis in this study suggests that the effect of body composition on shock attenuation may vary by sex. However, the body composition variables measured were limited to mass-related tissue properties. Other factors likely influence mechanical energy dissipation via tissue deformation, such as tissue architecture (e.g. stiffness, size) and geometry which tend to differ between sexes (Besson et al., 2022; Bruce & Edwards, 2023; Fouré et al., 2012; Meardon et al., 2021; Morse, 2011; Saeki et al., 2019). While the link between externally measured shock attenuation and internal tissue deformation still needs to be established, the present study motivates future explorations of the role of tissue architecture and geometry alongside tissue masses in impact-related energy dissipation during running, specifically considering the effects of sex on these associations.

5. Limitations

An interaction between gait mechanics, musculoskeletal tissue architecture and morphology which affect the degree of shock attenuated by active and passive mechanisms likely influenced the findings of this study. Additionally, there is limited evidence explaining what proportion of impact shock or which signal frequencies are attenuated by passive or active mechanisms, and this association cannot be determined with this study's methodology. Although shock attenuation in the frequency domain provides insights into the body's response to impact, it pertains to external mechanical loading. Therefore, inferring tissue-level stress and strain responses to impact shock and shock attenuation analyses alone is challenging. A large-sample study incorporating motion capture with accelerometry during gait, tissue imaging, electromyography, and computational tissue modeling may reveal the interaction between gait mechanics, musculoskeletal tissue architecture, and morphology for which the present study provides evidence for directional hypotheses.

6. Conclusions

Sex may not influence high-frequency (9–50 Hz) shock attenuation during running when tibial shock signal power is similar between sexes. However, based on this study's post-hoc analysis, body composition may differentially affect how female and male runners respond to shock attenuation demands. Thus, a larger sample is required to determine if the effect of body composition on shock attenuation varies by sex. Such sex differences may have implications for prospective running-related injury risk, though internal tissue stresses and strains occurring from passive shock attenuation must be quantified with computational models to help explain varying injury rates between male and female runners.

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CRediT authorship contribution statement

Gauri A. Desai: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Alexandra F. DeJong Lempke:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **Jaroslaw Harezlak:** Writing - review and editing, Methodology, Investigation, Formal Analysis. **Allison H. Gruber:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

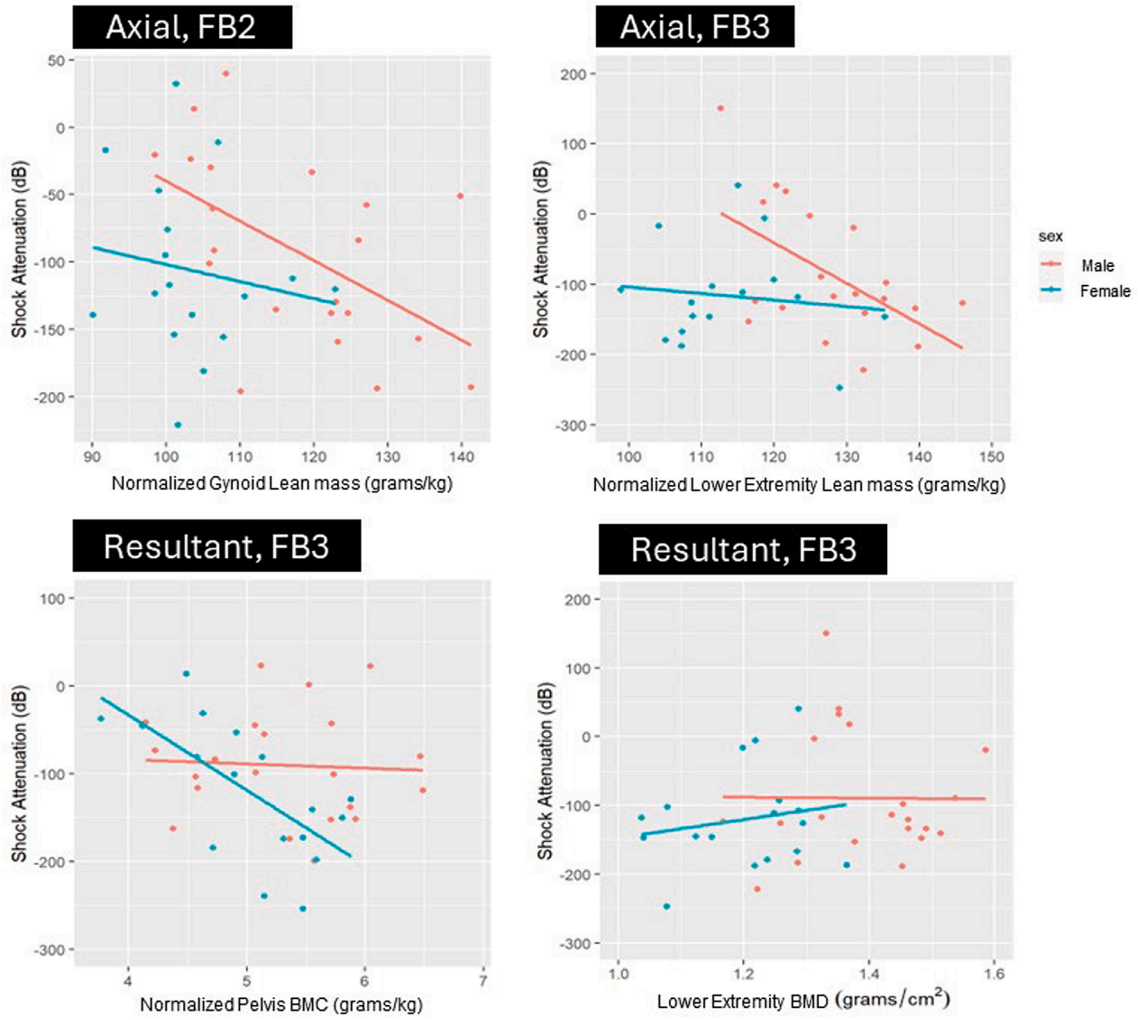


Fig. 5. Incorporating body composition (normalized to body mass) into the original model revealed previously insignificant associations between sex and shock attenuation becoming significant. Although the sample size is insufficient for a sex-stratified analysis, these visualizations were generated to better understand how body composition might affect the main effect of sex on shock attenuation. Only relationships where body composition appears to be related to shock attenuation differently in males versus females are presented here.

the work reported in this paper.

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Appendix A. Supplementary data

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